

ML2 - HW4

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Discussion

Q1. Model Architecture and Training procedure

Model and hyperparameters:

We chose to use GAN model. We trained an unconditional GAN on 64×64 RGB images with latent dimension 128. I used `BCEWithLogitsLoss` and Adam with learning rate $2 \cdot 10^{-4}$, $(\beta_1, \beta_2) = (0.5, 0.999)$, and batch size 64. Real images were normalized to $[-1, 1]$ to match the generator's `tanh` output.

Stability:

GAN training was not monotonic. Instability was stronger when the discriminator became too confident, so I used real-label smoothing for the discriminator (0.9 instead of 1.0), which made training more stable. I did not observe a clear persistent mode collapse (the generator did not produce the same sample repeatedly).

Q2. Training Loss Plots and Convergence

The generator and discriminator losses are not monotonic and show oscillations, which is typical for GANs. In our plots the losses remain bounded (no divergence), and sample quality improves over training, suggesting a stable training regime rather than strict convergence.

Q3. Conclusions

Our GAN learned to generate flower-like images, with noticeably better structure after longer training. The training was stable in practice (but didn't directly converge) and label smoothing helped when the discriminator became too confident. In addition, the similar/dissimilar pairs showed that visual similarity does not always correspond to similar latent vectors z , suggesting that the generator's mapping from z to images is complex and non-linear, so small changes in z can lead to noticeable visual changes. .

Q4. Generated Images

The generated samples generally look flower-like: they capture typical colors, green backgrounds, and coarse petal/flower structures. However, many images remain slightly blurry and contain artifacts, so fine details are not always sharp. Overall, the quality is reasonable for 64×64 generation, and it improves with longer training (for one train we used 80 epochs instead of 50 and it looked better detailed - but didnt improve the losses).

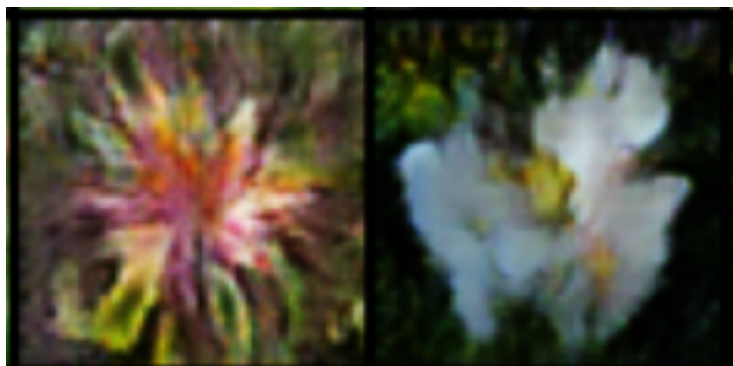


Figure 1: Generated samples from the trained GAN.

*Note that the images look here much blurrier than they really are because we enlarged them.

Q5. Latent Space Visualization

Based on the plotted z vectors and the measured L_2 distances, similar-looking images did not consistently have smaller latent distances than dissimilar images. Some visually similar pairs still had relatively large $L_2(z)$, and dissimilar pairs were not always farther apart. This is expected because the generator induces a complex, highly non-linear mapping from z to image space, so visual similarity does not necessarily correspond to simple Euclidean proximity in the latent space.